air batteries with practical specific energy ranging from 500 to 900Wh kg^{-1}, which is at least two to three times compared to those of alternative rechargeable batteries, and therefore would be sufficient to deliver a driving range of about 600km in one charge.

The limited performance of lithium-air batteries is the result of several technical barriers, in particular the architecture of the carbon cathode material and the catalyst. An ideal cathode should have a porous structure, where microporous channels can facilitate rapid oxygen diffusion, interconnected pore channels can act as reactive sites between lithium ions and oxygen, and mesopores might serve as the accommodation sites for discharge products. Carbon in the following forms is best suited for this purpose (see Table 2).

TABLE 2
ARCHITECTURE OF CARBON CATHODE

	Surface Area (m^2 gm^{-1})	Pore Dia (nm)	Specific Capacity (mAh^{-1})
Super-P	62	50	1736
Carbon nano balls	314	14.3	3490
Mesoporous graphene	-	up to 100	15,000
CNT (carbon nanotube)	-	15 to 30	2540

Due to its very high specific capacity, mesoporous graphene has received worldwide attention as a future cathode material.

The discharge and charging potential of lithium-air batteries are 2.7V and about 4.0V, respectively. Unfortunately, carbon materials are unstable above 3.5V in presence of Li_2O_2, forming Li_2CO_3, and severely reduce discharge-charge cycles. So, another main challenge is to design and use proper bifunctional catalysts that might give better ORR and OER activities to improve the cycle's performance. The design of the catalysts is yet to be perfected.

Overview

Presently a number of aprotic electrolyte solvents, namely, sulfonamide based electrolyte, ionic liquid, ethers, propylene carbonate/lithium bis-trifluoromethansulfonilimide, sulphones, etc have been recommended. However, electrolyte degradation during the lithium-air battery cycling process hampers their commercialisation. Moreover, some of the electrolytes, being inflammable, pose a safety hazard. Thus, developing a stable and safe electrolyte is the key to the improvement of aprotic batteries.

One of the major advantages of the aqueous lithium-air batteries is that the discharge reaction product is soluble in water, eliminating formation of electrically-resistive products that may passivate the air electrode. During discharge process, water-soluble LiOH is produced, but precipitates to form LiOH-H_2O crystals above its solubility limit of 5,25M. These crystals fill up the pores in the fluffy carbon electrode and retard the kinetics of both ORR and OER. But crucially the LiOH crystals do not coat and block the vital carbon surface that is generating the supply of voltage.

One way to overcome this problem is to use lithium iodide as 'facilitator' and water as co-reactant that boosts the lithium-air battery's capacity. It turns out that negatively charged iodide ions are converted into I_3 (triiodide) ions that combine with LiOH crystals and dissolve, allowing complete recharge by clearing the pores. In fact, this mechanism is even more effective than the recharge of Li_2O_2 attached to the electrode surface, since the electrons need not travel through Li_2O_2 layer, and hence less energy would be required to recharge the battery, promising an energy efficiency of 90% for aqueous lithium-air batteries.

Another disadvantage of the aqueous system is that its specific energy density is reduced significantly due to inclusion of solid electrolyte separator, which has considerable weight and volume.

The solid-state liquid-air battery is attractive with respect to the safety issues and long-term stability. However, lithium-stable and lithium dendrite formation-free high lithium ion conducting solid electrolytes are yet to be developed. Therefore, a lithium-stable interlayer—such as a liquid or polymer electrolyte—between the lithium anode and the solid electrolyte is required. The architecture of air electrode is very crucial as the cell capacity is dependent on it. Several other challenges still exist, such as:

1. Insufficient ionic conductivity of the solid electrolyte, which is much lower than of the organic liquid electrolyte. Conductivity can be enhanced by increasing the crystallinity and forming nanosized solid interface through the space charge layer effect.

2. Poor interfacial compatibility between electrode and electrolyte. High solid-solid interfacial resistances are usually produced after the initial charging process arising from large volume changes of the electrodes during lithiation/DE lithiation. These interfacial resistances can be reduced by depositing ultrathin layer of Al_2O_3 on garnet electrolyte. Another route to reduce the impedance is to increase the operating temperature, enabling speeding up of motion of ions in the solid-state system.

3. The sharp interphases may cause lamination or large strain during cycles. Rational design of functional grade in the electrode/electrolyte interface and optimisation of electrode materials are still lacking.

All these findings reveal a promising way forward for lithium-air technology, at a time when many other research groups have given up further research. As more researchers return to the subject following many recent breakthroughs, perhaps a commercial lithium-air battery will finally become a reality within two years. EFY